

- Q.30 Establish a relationship between Mass density and weight density.
- Q.31 Explain any five properties of fluids.
- Q.32 Draw a simple hydraulic circuit labelling all the components.
- Q.33 Sketch and explain basic pneumatic circuits in brief.
- Q.34 Explain Pressure ratings and selection criteria of hydraulic pipes.
- Q.35 Explain any one Control valve.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the different rules and norms followed for designing hydraulic and pneumatic circuits.
- Q.37 Write short note on followings:
- Air Motor
 - Air lubricator
- Q.38 Explain the construction and working of a Reciprocating pump with a neat sketch.

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4th Semester/ Mechatronics Subject: Hydraulic & Pneumatic Systems

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Valves are mainly
- Direction control valves
 - Pressure control valves
 - Flow control valves
 - All of the above
- Q.2 Which one of the following is a type of actuator in a hydraulic system?
- Pump
 - Valve
 - Strainer
 - Cylinder
- Q.3 The scientific principle that makes hydraulic systems possible is
- Boyle's law
 - Pascal's Principle
 - Continuity Equation
 - Bernoulli's principle
- Q.4 Which among the following is not the component of the FRL unit?
- Air filter
 - Air dryer
 - Air regulator
 - Air lubricator

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- Q.5 In which type of system does power transmission take place through compressed air?
 a) Fluid power system b) Hydraulic system
 c) Pneumatic system d) Stepper motors
- Q.6 The lubricator in a pneumatic circuit is the:
 a) First element in line b) Second element in line
 c) Last element in line d) Third element in line
- Q.7 The equation of continuity is
 a) $A/V = \text{Constant}$ b) $A \cdot V = \text{Constant}$
 c) $V/A = \text{Constant}$ d) $A \cdot \dot{V} = \text{Constant}$
- Q.8 The specific gravity of water is
 a) 1 b) 0
 c) 9.81 d) None of the above
- Q.9 What is the function of the air compressor?
 a) Decreases pressure of air
 b) Increases the pressure of air
 c) Removes dust particles
 d) Adds Lubricating oil
- Q.10 The Bernoulli equation is based on the principle of conservation of
 a) Mass b) Force
 c) Momentum d) Energy

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Full form of FRL is _____.
 Q.12 Define Mass Density.

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- Q.13 Priming is necessary for a _____ pump.
 Q.14 Centrifugal pumps are most commonly used when large discharge is required. (True/ False)
 Q.15 _____ is used to convert the power of hydraulic pressurized oil into mechanical energy.
 Q.16 In pneumatic system, medium used is _____.
 Q.17 Name any two types of Pneumatic Valves.
 Q.18 Write the function of Air Compressor.
 Q.19 Define Pascal's Law.
 Q.20 Define Hydrostatics.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a note on pipe material, types of fittings and connectors in the fluid system.
 Q.22 Explain Rotary type air compressor.
 Q.23 State Bernoulli's theorem with application.
 Q.24 Write different applications of pneumatic system in engineering field.
 Q.25 What is pump cavitation and what is its cause?
 Q.26 What is an air vessel? What are the functions of air vessel in suction and delivery sides in a reciprocating Piston pump?
 Q.27 Explain Pascal's law with the help of practical applications of it.
 Q.28 Compare the hydraulic system and pneumatic system.
 Q.29 State and explain the continuity equation.

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